

Ruichen Li

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Statement

Third-year Ph.D. student working on low-Earth orbit (LEO) satellite networks. My research interests include Earth-observation data processing, LEO satellite networking, space data center design, and agentic designs.

Education

The Ohio State University

Ph.D. in Computer Science and Engineering

- **Research Interests:** Mobile agents and LEO satellites.

Columbus, OH

Aug 2024 - present

University of California, San Diego

MS in Computer Science and Engineering

- GPA 4.00/4.00

San Diego, CA

Jan 2023 - Jun 2024

University of California, San Diego

BS in Computer Science

- GPA 3.91/4.00, top 5% in Computer Science and Engineering department and graduated with Provost's Honors

San Diego, CA

Sep 2019 - Dec 2022

Research Experience

LEO Satellite Research

The Ohio State University

- Proposed and developed onboard storage optimization techniques in Earth observation (EO) satellite constellations, the related paper was accepted by **SIGCOMM 2026**
- Proposed multi-service processing techniques for EO constellations and one publication is currently under review
- Implemented and experimented LEO network scheduling system for continuous monitoring and dynamic tracking, with one publication accepted by **SenSys 2026**

Columbus, OH

Aug 2024 - present

Agent Research

The Ohio State University

- Developed mobile agent safety benchmark and one publication is currently under review.

Columbus, OH

Aug 2024 - Dec 2025

Information Alignment Model Research

University of California, San Diego

- Developed a holistic information alignment evaluation metric for LLMs.
- AlignScore paper was accepted by **ACL 2023** and received **best paper award** at DialDoc@ACL 2023, introducing a SOTA level holistic metric on factual consistency
- Subsequent paper accepted by **NeurIPS 2023**, showing its efficiency and performance on diverse information alignment tasks

San Diego, CA

Jun 2022 - Dec 2023

Work Experience

Goertek Inc.

Research Intern

- Developed a **hand gesture recognition system for VR**, achieving over 90% accuracy using a single IMU-based smart ring
- Implemented linear discriminant analysis and support vector machine API in C for the CMSIS-DSP platform
- Researched hand gesture feature engineering and CNN/GAN methods for a combined hand gesture recognition using IMU data

Beijing, China

Jul 2023 - Aug 2023

The Ohio State University

CSE Teaching Assistant

- TA for Introduction to Computer-Assisted Problem Solving for Construction Systems Management

Columbus, OH

Aug 2024 - Present

University of California, San Diego

CSE Teaching Assistant

- Experienced TA for Discrete Math and Mathematics for Algorithms with Russell Impagliazzo and Miles Jones as instructors
- Primary responsibilities include conducting office hours, grading assignments / exams, and organizing staff meetings

San Diego, CA

Sep 2021 - Jun 2024

Publications

Achieving Efficient Storage and Communication via Collaboration

Ruichen Li, Yufan Wu, Zhengyi Hu, Sheng-Jyun Cai, Laang Wei, Qifan Yang, Ting Zhu
SIGCOMM, 2026

Exploring LEO Satellite Networks for Continuous Monitoring and Dynamic Tracking

Lang Wei, **Ruichen Li**, Baodong Chen, Qifan Yang, Yufan Wu, Ting Zhu
SenSys, 2026

AlignScore: Evaluating Factual Consistency with a Unified Alignment Function

Yuheng Zha, Yichi Yang, **Ruichen Li**, Zhiting Hu
ACL, 2023

Text Alignment is an Efficient Unified Model for Massive NLP Tasks
Yuheng Zha, Yichi Yang, **Ruichen Li**, Zhiting Hu
NeurIPS, 2023

Service _____

Reviewer ICLR 2024, ICML 2026, NeurIPS 2026, TMLR 2026, AAAI 2027

Artifact Evaluation Usenix Security 2025, MobiSys 2025

Awards & Achievements _____

2026	College of Engineering Travel Award , \$1,000 for travel grant	<i>Columbus, OH</i>
2023	Best Paper Award , DialDoc@ACL	<i>San Diego, CA</i>

Project Experience _____

Pomodoro Timer, link to website	<i>Remote</i>
Software Engineer / Quality Assurance Engineer	<i>Jan 2021 - Mar 2021</i>

- Building task-oriented timer with work/break intervals and a to-do list remaindering the user of the finished and upcoming tasks
- Developed with HTML, CSS, and JavaScript for its timer and to-do list functions
- Unit testing with Jest and end-to-end testing with Cypress with 100% coverage

Skills _____

Programming	Python (Pandas, PyTorch, NumPy, Scikit-learn. etc.), Java, R(ggplot2), C/C++, HTML/CSS, JavaScript(Node.js)
Miscellaneous	Linux, Shell(Bash/Zsh), LaTeX(Overleaf/Markdown), Git, MongoDB, Figma, Notion
Languages	Mandarin(Native), English(Fluent), Japanese(Basic)